SAKILA PROJECT

1. TITLE :

Sakila SQL Business Analysis Project

2. INTRODUCTION :

- Sakila is a movie rental database.
- It contains data about customers, movies, rentals, payments, stores and staff.
- It represents a real-world rental business system.

3. OBJECTIVE :

- Analyze customer behavior, rental trends, and revenue patterns using SQL
- Extract meaningful business insights from relational data
- Identify top customers, popular movies, and high-performing categories
- Analyze revenue distribution across stores and categories
- Study monthly rental and revenue trends to understand seasonality
- Provide data-driven insights to support business decision-making

4. TABLES USED :

The analysis is performed using multiple relational tables from the Sakila database, including customers, rentals, payments, films, inventory, and categories. These tables are connected using primary and foreign keys, enabling complex joins to extract meaningful business insights.

TABLE	DESCRIPTION
customers	Customer demographic details
rental	Rental transaction records
payment	Payment and revenue data
film	Movie information
inventory	Store inventory tracking
category	Movie category classification
staff	Staff managing transactions
store	Store location and details

5. TOOLS USED

- MySQL Workbench — SQL Queries
- Sakila Database — Dataset
- SQL — Data Analysis

6. BUSINESS QUESTIONS

1. How many customers does the business have?
2. How many total rentals were made?
3. What is the total revenue generated?
4. Which store generates the most revenue?
5. Who are the top customers by total spending?
6. Which movies are rented the most?
7. Which movie categories generate the most revenue?
8. How many rentals happen each month?
9. How does revenue change over time?
10. Which actors have appeared in the most movies?
11. Which category has the most movies?
12. Which categories are rented the most?

To answer the business questions, SQL queries were written and the results were analyzed to generate business insights.

7. ANALYSIS

I. How many customers does the business have?

```
-- Total customers

SELECT COUNT(*) AS total_customers
FROM customer;
```

RESULT :

total_customers
599

INSIGHTS : The customer base consists of 599 users, indicating moderate scale and potential for growth through targeted acquisition strategies. This shows the size of the customer base and helps understand how many customers are using the rental service.

II. How many total rentals were made?

```sql
-- Total rentals

SELECT COUNT(*) AS total_rentals
FROM rental;
```

RESULT :

total_rentals
16044

INSIGHTS : A total of 16,044 rentals indicate strong customer engagement and frequent usage of the platform, which shows that customers frequently rent movies. This indicates that the business has good customer activity and engagement.

III. What is the total revenue generated?

```sql
-- Total revenue

SELECT sum(amount) AS total_revenue
FROM payment;
```

RESULT :

total_revenue
67406.56

INSIGHTS : The business generated total revenue from all rentals, which shows the overall earnings of the company. Revenue is an important measure to understand business performance.

IV. Which store generates the most revenue?

```sql
-- Revenue by Store

SELECT staff_id, SUM(amount) AS revenue
FROM payment
GROUP BY staff_id;
```

RESULT :

staff_id	revenue
1	33482.50
2	33924.06

INSIGHTS : The revenue is generated by two stores, and we can compare which store performs better. This helps the business understand store performance and manage stores more efficiently.

V. Who are the top customers by total spending?

```sql
-- Top 10 Customers with Name

SELECT c.customer_id,
       c.first_name,
       c.last_name,
       SUM(p.amount) AS total_spent
FROM payment p
JOIN customer c
    ON p.customer_id = c.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY total_spent DESC
LIMIT 10;
```

RESULT :

customer_id	first_name	last_name	total_spent
526	KARL	SEAL	221.55
148	ELEANOR	HUNT	216.54
144	CLARA	SHAW	195.58
137	RHONDA	KENNEDY	194.61
178	MARION	SNYDER	194.61
459	TOMMY	COLLAZO	186.62
469	WESLEY	BULL	177.60
468	TIM	CARY	175.61
236	MARCIA	DEAN	175.58
181	ANA	BRADLEY	174.66

INSIGHTS : The top customers contribute a large portion of total revenue. These customers are very important for the business and can be targeted for loyalty programs and special offers.

VI. Which movies are rented the most?

```sql
-- Most rented movies

SELECT f.title, COUNT(*) AS rental_count
FROM rental r
JOIN inventory i
    ON r.inventory_id = i.inventory_id
JOIN film f
    ON i.film_id = f.film_id
GROUP BY f.title
ORDER BY rental_count DESC
LIMIT 10;
```

RESULT :

title	rental_count
BUCKET BROTHERHOOD	34
ROCKETEER MOTHER	33
RIDGEMONT SUBMARINE	32
GRIT CLOCKWORK	32
SCALAWAG DUCK	32
JUGGLER HARDLY	32
FORWARD TEMPLE	32
HOBBIT ALIEN	31
ROBBERS JOON	31
ZORRO ARK	31

INSIGHTS : High-demand movies drive rental frequency, suggesting the importance of maintaining sufficient inventory for popular titles, which means they are the most popular among customers. The business should keep more inventory of these popular movies.

VII. Which movie categories generate the most revenue?

```sql
-- Revenue by Category

SELECT c.name, SUM(p.amount) AS revenue
FROM payment p
JOIN rental r
    ON p.rental_id = r.rental_id
JOIN inventory i
    ON r.inventory_id = i.inventory_id
JOIN film_category fc
    ON i.film_id = fc.film_id
JOIN category c
    ON fc.category_id = c.category_id
GROUP BY c.name
ORDER BY revenue DESC;
```

RESULT **:**

name	revenue
Sports	5314.21
Sci-Fi	4756.98
Animation	4656.30
Drama	4587.39
Comedy	4383.58
Action	4375.85
New	4351.62
Games	4281.33
Foreign	4270.67
Family	4226.07
Documentary	4217.52
Horror	3722.54
Children	3655.55
Classics	3639.59
Travel	3549.64
Music	3417.72

INSIGHTS : Some movie categories generate more revenue than others. The business should focus more on high-revenue categories to increase profit.

VIII. How many rentals happen each month?

```sql
-- Rentals per Month

SELECT DATE_FORMAT(rental_date, '%Y-%m') AS month,
       COUNT(*) AS total_rentals
FROM rental
GROUP BY month
ORDER BY month;
```

RESULT :

month	total_rentals
2005-05	1156
2005-06	2311
2005-07	6709
2005-08	5686
2006-02	182

INSIGHTS : The number of rentals changes from month to month, showing that customer demand is not constant throughout the year.
Some months have higher rentals, which may be due to holidays, weekends, or seasonal demand.

IX. How does revenue change over time (monthly trend)?

```sql
-- Monthly revenue trend

SELECT DATE_FORMAT(payment_date, '%Y-%m') AS month,
       SUM(amount) AS revenue
FROM payment
GROUP BY month
ORDER BY month;
```

RESULT :

month	revenue
2005-05	4823.44
2005-06	9629.89
2005-07	28368.91
2005-08	24070.14
2006-02	514.18

INSIGHTS : Revenue changes over time, and some months have higher revenue than others. This helps the business understand seasonal trends and plan marketing strategies.

X. Which actors have appeared in the most movies?

```sql
-- Top actors

SELECT a.actor_id,
       a.first_name,
       a.last_name,
       COUNT(fa.film_id) AS movie_count
FROM actor a
JOIN film_actor fa
ON a.actor_id = fa.actor_id
GROUP BY a.actor_id, a.first_name, a.last_name
ORDER BY movie_count DESC
LIMIT 10;
```

RESULT :

actor_id	first_name	last_name	movie_count
107	GINA	DEGENERES	42
102	WALTER	TORN	41
198	MARY	KEITEL	40
181	MATTHEW	CARREY	39
23	SANDRA	KILMER	37
81	SCARLETT	DAMON	36
37	VAL	BOLGER	35
158	VIVIEN	BASINGER	35
106	GROUCHO	DUNST	35
13	UMA	WOOD	35

INSIGHTS : These actors have appeared in the highest number of movies, which means they are frequently featured in the film inventory.
The business can use movies featuring these popular actors for promotions to attract more customers and increase rentals.

XI. Which category has the most movies?

```sql
-- Movies per category

SELECT c.name,
       COUNT(fc.film_id) AS number_of_movies
FROM film_category fc
JOIN category c ON fc.category_id = c.category_id
GROUP BY c.name
ORDER BY number_of_movies DESC;
```

RESULT :

name	number_of_movies
Sports	74
Foreign	73
Family	69
Documentary	68
Animation	66
Action	64
New	63
Drama	62
Games	61
Sci-Fi	61
Children	60
Comedy	58
Classics	57
Travel	57
Horror	56
Music	51

INSIGHTS : These categories are rented the most, which shows customer preferences. The business should focus more on these popular categories and keep more inventory in these categories.

XII. Which categories are rented the most?

```sql
-- Top 5 Categories by Number of Rentals

SELECT c.name,
       COUNT(*) AS total_rentals
FROM rental r
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN film_category fc ON i.film_id = fc.film_id
JOIN category c ON fc.category_id = c.category_id
GROUP BY c.name
ORDER BY total_rentals DESC
LIMIT 5;
```

RESULT :

name	total_rentals
Sports	1179
Animation	1166
Action	1112
Sci-Fi	1101
Family	1096

INSIGHTS : The categories with the most movies have a larger inventory, which means the business offers more variety in those categories.
Having more movies in a category can attract more customers because customers have more choices to rent from.

8. CONCLUSION :

In this project, we analyzed the Sakila movie rental database using SQL to understand business performance and customer behavior.
We analyzed total customers, rentals, and revenue to understand overall business size.
We identified top customers, most rented movies, and most popular categories to understand customer preferences.
We analyzed revenue by store and category to understand business performance.
We also analyzed monthly rentals and revenue trends to understand how business changes over time.
Additionally, we analyzed actors and movie categories to understand movie distribution and popularity.
This project shows how SQL can be used to analyze business data and generate useful insights for decision-making.